

Implementation of the Rhie & Chow Interpolation

In this document, I will be going over the Rhie & Chow Interpolation method and its implementation into the accompanying code in the [Github repo](#).

This document will cover what the problem being addressed, the solution (Rhie & Chow Interpolation), and how it is implemented into the code.

I learned from and really got all the information in this document from the following lectures (which I highly recommend):

- [\[CFD\] Rhie & Chow Interpolation \(Part 1\): Chequerboard Oscillations](#)
- [\[CFD\] Rhie & Chow Interpolation \(Part 2\): Staggered Grids](#)
- [\[CFD\] Rhie & Chow Interpolation \(Part 3\): Deriving the Correction!](#)

Honestly, this document is more for me so that I can keep track of the implementation for debugging (which helped a lot since I referred to this often to resolve my errors).

I also refer to something called the “main” document which is the *SIMPLE Algorithm using FVM for 2D Incompressible N-S Eqns* pdf.

Okay, let’s begin!

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The Problem

In order to understand Rhie & Chow interpolation, we need to first understand what the problem being addressed is. I will be introducing the problem in a somewhat vaguely similar manner to the YouTube video in the first page,

First let us begin from the starting point of the momentum equation in it discretized matrix form as in Equation (1) which is Equation (7) in the “main” document.

$$U^* = \mathcal{A}^{-1}(\mathcal{H} - \nabla p) \quad (1)$$

In the equation above, $\mathcal{H}$ is the off-diagonal terms that represents the contribution from the neighboring cells and $-\nabla p$ is the pressure gradient across the face (the negative being there since a negative gradient (high to low) should be a positive contribution to the momentum).

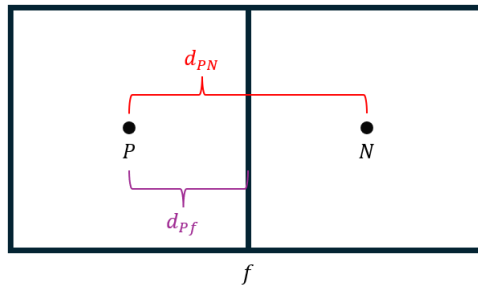
This is the first step for our segregated SIMPLE algorithm which solves the momentum equations first for the predicted velocity (U^*).

The next step is computing the pressure correction (p'). This is where our issues begin! The equation for the pressure correction is the pressure Poisson equation as in Equation (2) which is Equation (15) from the “main” document.

$$\nabla \cdot (\mathcal{A}^{-1} \nabla p') = \nabla \cdot U^* \quad (2)$$

At this step, we need to evaluate $\nabla \cdot U^*$ which we evaluate by taking the integral then applying the divergence theorem to get the integral as the summation of $(U^*)_f \cdot \mathbf{n}_f S_f$ over all the faces (covered in Equation (26)-(27) of the “main” document).

In order to get the value of $(U^*)_f$, we use linear interpolation of the cell center values as in Equation (3) below using the notation in the diagram below. Note the diagram below is for a 1D uniform mesh case (though it's drawn as a 2D cell) which we will assume for simplicity.



$$(U^*)_f = (l_x)(U^*)_P + (1 - l_x)(U^*)_N \quad (3)$$

$$\text{where } l_x \rightarrow d_{Pf}/d_{PN}$$

Nothing is wrong at this point. We have simply conducted a linear interpolation. Harmless right?

The following step is the updating of the solution. Specifically, let us consider the update for the velocity from the pressure correction as in Equation (4) which is Equation (18) in the “main” document.

$$U^{(k+1)} = U^{(k)} + \alpha_U \left((U^* - \mathcal{A}^{-1} \nabla p') - U^{(k)} \right) \quad (4)$$

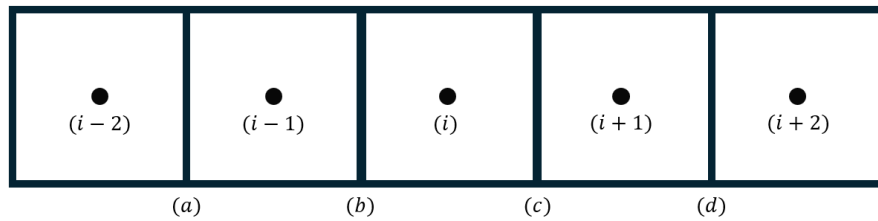
We have all the terms in Equation (4) already defined in a vector or matrix that we can apply the operation on except for $\nabla p'$. We already have the pressure correction (p'), so we just need to apply the gradient which we can do using a Gauss-gradient as in Equation (5) which we derived in the steps of Equation (45) – (47) of the “main” document.

$$\nabla p' = \frac{1}{A} \sum_f^{faces} \left(p'_f n_{fx} + p'_f n_{fy} \right) S_f \quad (5)$$

The need for interpolation here comes from needing the face values of the pressure correction (p'_f). So again, considering the same diagram and same linear interpolation we get the following Equation (6).

$$(p')_f = (l_x)(p')_P + (1 - l_x)(p')_N \quad (6)$$

So those are the steps and the linear interpolations used, but why does that become a problem? Let's consider a 1D example based on the diagram below to see how odd-even decoupling comes out from the linear interpolations.



Considering that we are interested in finding velocity at cell i , let us start with finding the predicted velocity at that cell which would be according to Equation (1) which is shown again below.

$$U^* = \mathcal{A}^{-1}(\mathcal{H} - \nabla p) \quad (1)$$

The next step is the pressure correction as in Equation (2) which is shown again below.

$$\nabla \cdot (\mathcal{A}^{-1} \nabla p') = \nabla \cdot U^* \quad (2)$$

Recall that for our finite volume method approach we take the integral then apply the divergence theorem as in Equation (7) and (8). In 1D, this simply becomes the difference between the east and west face as shown in Equation (9).

$$\int_{CV} \nabla \cdot (\mathcal{A}^{-1} \nabla p') dx = \int_{CV} \nabla \cdot U^* dx \quad (7)$$

$$\sum_{faces} (\mathcal{A}^{-1} \nabla p')_f = \sum_{faces} (U^*)_f \quad (8)$$

$$(\mathcal{A}^{-1} \nabla p')_{east} - (\mathcal{A}^{-1} \nabla p')_{west} = (U^*)_{east} - (U^*)_{west} \quad (9)$$

For simplicity, let us rewrite the LHS as a discrete matrix operator multiplied by the pressure correction. On the RHS let us substitute the interpolation to the face as in Equation (3) using the notation in the diagram. Let us also just consider this for an arbitrary cell i for simplicity.

$$(\mathcal{A}^{-1} \nabla p')_c - (\mathcal{A}^{-1} \nabla p')_b = (U^*)_c - (U^*)_b \quad (10)$$

$$\mathcal{L}_i p'_i = \left[\left(\frac{d_{i,c}}{d_{i,i+1}} \right) (U^*)_{i+1} + \left(1 - \frac{d_{i,c}}{d_{i,i+1}} \right) (U^*)_i \right] - \left[\left(\frac{d_{i-1,b}}{d_{i-1,i}} \right) (U^*)_{i-1} + \left(1 - \frac{d_{i-1,b}}{d_{i-1,i}} \right) (U^*)_i \right] \quad (11)$$

where $\mathcal{L}_i \rightarrow$ discrete matrix operator for pressure gradient

We can slightly simplify this by grouping terms as follows in Equation (12).

$$p'_i = \mathcal{L}^{-1} \left[\left(\frac{d_{i,c}}{d_{i,i+1}} \right) (U^*)_{i+1} - \left(\frac{d_{i-1,b}}{d_{i-1,i}} - \frac{d_{i,c}}{d_{i,i+1}} \right) (U^*)_i - \left(\frac{d_{i-1,b}}{d_{i-1,i}} \right) (U^*)_{i-1} \right] \quad (12)$$

It is important to note at this point that if the grid is uniform, then the coefficient in front of the $(U^*)_i$ goes to zero and the coefficient in front of the $(U^*)_{i+1}$ and $(U^*)_{i-1}$ go to half (consider why this is the case using the diagram on the previous page if this isn't intuitive to you). This would, for a uniform mesh result in Equation (13).

$$p'_i = \mathcal{L}^{-1} \frac{[(U^*)_{i+1} - (U^*)_{i-1}]}{2} \quad (13)$$

(for a uniform mesh)

Notice that this is just a central difference and that there is no dependence on the $(U^*)_i$. It is equally important to realize that the coefficient in front of $(U^*)_i$ is closer to zero the more uniform the mesh is and closer to 1 the more non-uniform it is.

$$\left(\frac{d_{i-1,b}}{d_{i-1,i}} - \frac{d_{i,c}}{d_{i,i+1}} \right) \text{ goes to 0 the more uniform the mesh is}$$

$\left(\frac{d_{i-1,b}}{d_{i-1,i}} - \frac{d_{i,c}}{d_{i,i+1}}\right)$ goes to 1 the more non-uniform the mesh is

Therefore, if our mesh is only slightly non-uniform, we have only a slight dependence on $(U^*)_i$ since the leading coefficient is small. You can kind of think of it like a convex averaging where the weighting of $(U^*)_i$ is dependent on the uniformity or lack thereof.

Okay, so where the problem shows itself is when we next apply the pressure correction in order to update the velocity as in Equation (4) which is shown again below.

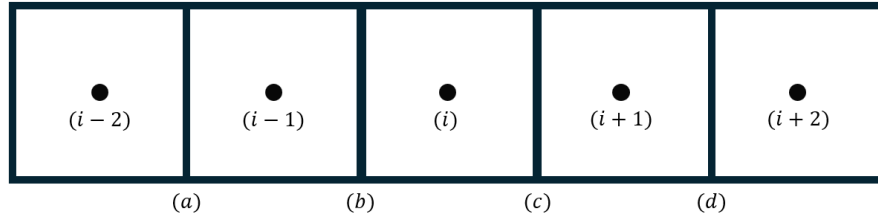
$$U^{(k+1)} = U^{(k)} + \alpha_U \left((U^* - \mathcal{A}^{-1} \nabla p') - U^{(k)} \right) \quad (4)$$

The issue comes in determining the value of $\nabla p'$ which, as aforementioned, we compute using Equation (5) as shown below and linearly interpolate the for the face values as in Equation (6) which is also shown again below.

$$\nabla p' = \frac{1}{A} \sum_f^{faces} (p'_f n_{fx} + p'_f n_{fy}) S_f \quad (5)$$

$$(p')_f = (l_x)(p')_P + (1 - l_x)(p')_N \quad (6)$$

Let us again consider this for a 1D case based on the diagram which is shown again below (I keep showing things again because I don't want to scroll up just by the way).



Evaluating Equation (5) considering the cell i in 1D and substituting Equation (6) to get the face values we get the following Equation (14).

$$\nabla p'_i = \frac{(p'_c - p'_b)}{d_{bc}} = \frac{1}{d_{bc}} \left[\left(\frac{d_{i,c}}{d_{i,i+1}} \right) (p')_{i+1} + \left(1 - \frac{d_{i,c}}{d_{i,i+1}} \right) (p')_i \right] - \left[\left(\frac{d_{i-1,b}}{d_{i-1,i}} \right) (p')_{i-1} + \left(1 - \frac{d_{i-1,b}}{d_{i-1,i}} \right) (p')_i \right] \quad (14)$$

Notice that Equation (14) takes virtually the same form as Equation (12) so the same ideas hold and, as such, for a uniform mesh we can write Equation (14) as Equation (15).

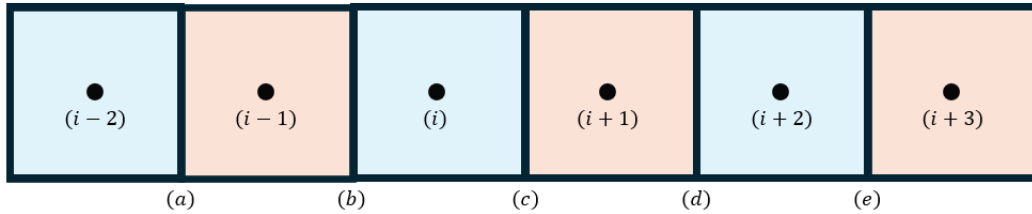
$$\nabla p'_i = \frac{[(p')_{i+1} - (p')_{i-1}]}{d_{bc}} \quad (15)$$

(for a uniform mesh)

For the sake of argument, let us just consider the uniform mesh scenario so that we can avoid expression swell and plug Equation (13) into Equation (15) which would give us Equation (16).

$$\nabla p'_i = \frac{\mathcal{L}_{i+1}^{-1} \frac{[(U^*)_{i+2} - (U^*)_i]}{2} - \mathcal{L}_{i-1}^{-1} \frac{[(U^*)_i - (U^*)_{i-2}]}{2}}{d_{bc}} \quad (16)$$

WHAT! This should be shocking... The pressure gradient, which is what is used to apply the pressure correction to the updated velocity, only depends on $(U^*)_{i+2}$, $(U^*)_i$, and $(U^*)_{i-2}$. A diagram is shown below where the cells highlighted in blue are the ones used to determine the solution at cell i and highlighted in orange are the ones used to determine the solution at cell $i + 1$. Notice the odd-even decoupling!



blue → cells used for computing pressure gradient for cell i

orange → cells used for computing pressure gradient for cell $i + 1$

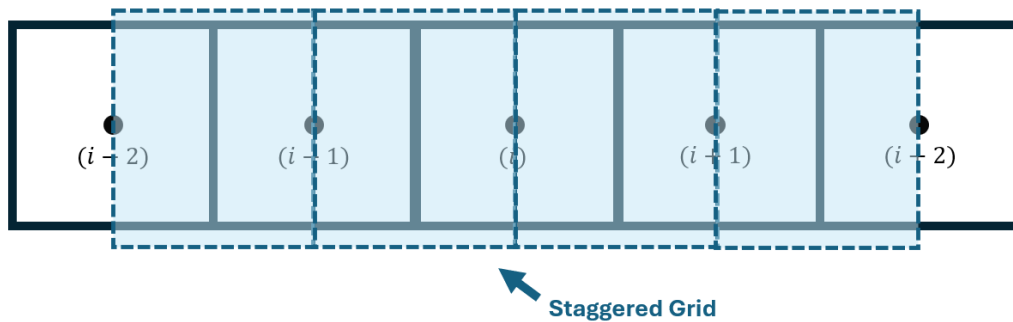
So, this is our problem. We essentially have two different solutions because the odd and even cells are decoupled. Note that this is a result of the discretization and utilization of linear interpolation, not inherently the uniformity of the mesh (though it plays a factor). For a non-uniform mesh, the coupling between the odd and even cells is dependent on how non-uniform as the closer the cell is to uniform, the weaker the coupling is (as explained on the previous page(s)). The uniform case is the only one where the odd and even cells are completely decoupled. Yet, weak coupling, and hence checkerboard oscillations, can still be seen on non-uniform grids. This, very briefly, is our problem: odd-even decoupling. Now onto ...

The Solution

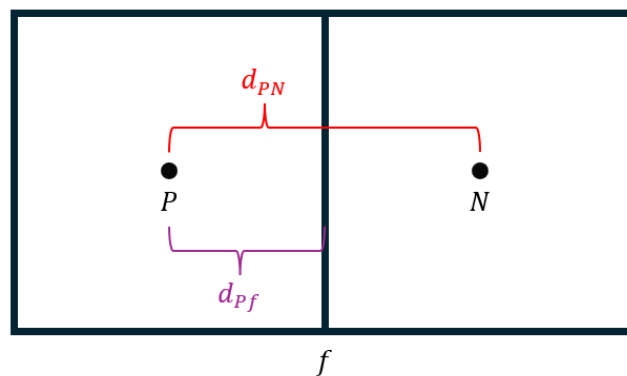
The solution is very simple. Only like a couple lines of code really but it fixes this problem entirely. The solution is using a Rhie & Chow interpolation instead of a linear interpolation for the pressure gradient. Let's go through the formulation of Rhie & Chow interpolation to see what this means.

If the compounding of linear interpolation was the issue, we need to not use linear interpolation in either the pressure correction or the solution update. To not use linear interpolation means that we need to find the face value through some other mechanism in either step.

What Rhie (Ford Motor Co.) and Chow (UIUC) thought to do is *find the face value of the predicted velocity in the pressure correction step by solving a momentum balance for a staggered cell at the face*. The idea comes naturally from staggered grids, which is what people used to use before this method came out, but we will not really be discussing staggered grids in detail. The general idea though is to have one grid for velocity that is staggered and another grid for everything else. The velocity is staggered so that the cell centers of the velocity grid are the faces of the “other” grid (as in the diagram below) so no interpolation is needed, we just use the direct value. The key point is that the staggered grid solved for the velocity at the face... can we do that without a staggered grid? Yes, and that is where Rhie & Chow interpolation comes from.



Recall Equation (1) as shown below and Equation (17) which rewrites Equation (1) for the face value of the predicted velocity. This is the equation we would want to solve to avoid interpolation. Consider also the following diagram from before which is shown again below.



$$U^* = \mathcal{A}^{-1}(\mathcal{H} - \nabla p) = \mathcal{A}^{-1}\mathcal{H} - \mathcal{A}^{-1}\nabla p \quad (1)$$

$$U_f^* = \mathcal{A}_f^{-1}\mathcal{H}_f - \mathcal{A}_f^{-1}\nabla p_f \quad (17)$$

If we can evaluate U_f^* we would not need to linearly interpolate for that face values as we did in Equation (11) which would consequently stop the sequence of events that cause the odd-even decoupling. How can we evaluate Equation (17)?

Well to make things simpler, we are going to introduce a notation that people often use and was used in the linked videos on the first page. Equation (18) below represents the momentum balance equation rearranged to get the value of the velocity at cell P .

$$U_P = -\frac{1}{a_P} \sum a_N U_N - \frac{1}{a_P} \nabla p_P = \tilde{U}_P - \frac{1}{a_P} \nabla p_P \quad (18)$$

$$\text{where } \frac{1}{a_P} \rightarrow \mathcal{A}_P^{-1}$$

$$\text{where } \mathcal{H}_P \rightarrow \sum a_N U_N$$

$$\text{where } -\frac{1}{a_P} \sum a_N U_N \rightarrow \tilde{U}_P$$

Specifically, for the $\mathcal{H}$ term, recall that it represents the influence of our neighbors, therefore we were able to write it as the summation of some coefficient representing the operator times the neighbor velocities. So now applying Equation (18) to Equation (17) we get Equation (19).

$$U_f^* = -\frac{1}{a_f} \sum a_f U_f - \frac{1}{a_f} \nabla p_f = \tilde{U}_f^* - \frac{1}{a_f} \nabla p_f \quad (19)$$

We can evaluate $\frac{1}{a_f}$ through a linear interpolation and we can evaluate ∇p_f by using the direct cell center values. How do we get $\sum a_f U_f$? We can get $\tilde{U}_f^*$ by interpolating between $\tilde{U}_P^*$ and $\tilde{U}_N^*$ then substitute everything back into Equation (19) to ultimately get Equation (23). Let's go through the math then I will explain it... (I will also explain as we go a little).

$$U_f^* = \tilde{U}_f^* - \frac{1}{a_f} \nabla p_f \rightarrow \tilde{U}_f^* = U_f^* + \frac{1}{a_f} \nabla p_f \quad (20)$$

So, $\tilde{U}_f^*$ just is the predicted velocity without the pressure gradient contribution i.e. $\tilde{U}_f^* \rightarrow$ pressure-free momentum (this is kind of important to understand).

We can evaluate $\tilde{U}_f^*$ through a linear interpolation. So we are interpolating pressure-free momentums in Equation (21).

$$\tilde{U}_f^* = l_x \tilde{U}_P^* + (1 - l_x) \tilde{U}_N^* = l_x \left(U_P^* + \frac{1}{a_P} \nabla p_P \right) + (1 - l_x) \left(U_N^* + \frac{1}{a_N} \nabla p_N \right) \quad (21)$$

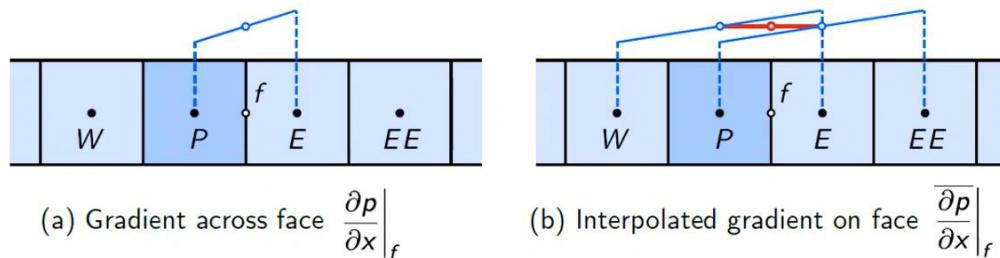
Plug Equation (21) back into (20) to get Equation (22) then rearrange and bracket terms to get Equation (23).

$$U_f^* = l_x \left(U_p^* + \frac{1}{a_p} \nabla p_p \right) + (1 - l_x) \left(U_N^* + \frac{1}{a_N} \nabla p_N \right) - \frac{1}{a_f} \nabla p_f \quad (22)$$

$$U_f^* = [l_x U_p^* + (1 - l_x) U_N^*] + \left[l_x \left(\frac{1}{a_p} \nabla p_p \right) + (1 - l_x) \left(\frac{1}{a_N} \nabla p_N \right) \right] - \left[\frac{1}{a_f} \nabla p_f \right] \quad (23)$$

Okay... that was a lot of math. Not difficult math, but a lot of it. Let's walk through what happened and what it means. We linearly interpolated for $\tilde{U}_f^*$ then used that in our evaluation of the face value U_f^* . We use a linear interpolation here. Is that a problem? No. We are interpolating the pressure-free momentum ($\tilde{U}_f^*$), but before the issue was interpolating the *full velocity which importantly included the pressure gradient*. We are not interpolating the pressure contribution.

Think about it this way. In Equation (23), the first term is the linear interpolation which we were doing before as in Equation (11). The second and third bracketed terms on the RHS is the correction to that interpolation! The second bracketed term on the RHS is the linear interpolation of the pressure gradient (that turns the momentum pressure free) and the third term is the direct pressure gradient at the face (the new correct pressure gradient that we use). The difference between the second and third term represents how different the linear interpolation is compared to the direct gradient and applies a correction based on that difference. An image taken from the [linked video](#) is shown that represents (a) as the second bracketed term on the RHS of Equation (23) and (b) as the third bracketed term on the RHS of Equation (23).



I'm kind of rambling, but here is another better way to look at it in my opinion. Equation (23) can be rewritten as Equation (24) below.

$$\tilde{U}_f^* = U_f^* + \left[\frac{1}{a_f} \nabla p_f \right] \quad (24)$$

$$\text{which comes from } \rightarrow U_f^* = \tilde{U}_f^* - \frac{1}{a_f} \nabla p_f$$

The $\tilde{U}_f^*$ is really just the momentum equation with the pressure gradient contribution subtracted. Then after algebraically removing the pressure gradient (by having subtracted the linear interpolation for $\frac{1}{a_f} \nabla p_f$), we instead add the direct face pressure gradient contribution. We do interpolate the pressure-free momentum ($\tilde{U}_f^*$) but we no longer use the interpolated pressure gradient as before which is the second bracketed term in Equation (23). That is to say for the pressure gradient we are using:

$$\frac{p_{i+1} - p_i}{a_f} \text{ for east contribution,} \quad -\frac{p_i - p_{i-1}}{a_f} \text{ for west contribution}$$

instead of

$$\frac{p_{i+1} + p_i}{2a_f} \text{ for east contribution,} \quad -\frac{p_i + p_{i-1}}{2a_f} \text{ for west contribution}$$

(for a uniform mesh)

The top row is the direct face pressure gradient instead of the linear interpolation we would do (as shown for a uniform mesh) on the bottom row. Notice that when you add the contributions, for the direct face pressure gradient and linear interpolation you get the following below.

$$\text{direct pressure gradient contributions} \rightarrow \frac{p_{i+1} - 2p_i + p_{i-1}}{a_f}$$

$$\text{linear interpolation contributions} \rightarrow \frac{p_{i+1} - p_{i-1}}{2a_f}$$

So, you can see clearly that for the direct pressure gradient there is a dependence on p_i but there is not one for the linear interpolation. That's the point! That is what we were trying to achieve! Yes, we did interpolate for the pressure free momentum, but that is not a problem. It may seem obvious to others, but I questioned as to why it was not a problem. It is not a problem because it is not a gradient at a cell center its just an average face value. Interpolating gradients at cell centers is the real issue since it requires a wide stencil instead of two adjacent cells. Interpolation in general is not a problem. And, in the corrected formulation, we did not interpolate the pressure gradient contribution to that momentum and instead use Rhie & Chow interpolation so we do have influence from the current cell. Therefore, no odd-even decoupling!

(this was kind of messy. I am not superbly proud of myself, but I am also tired. It makes sense to me... is that enough?)

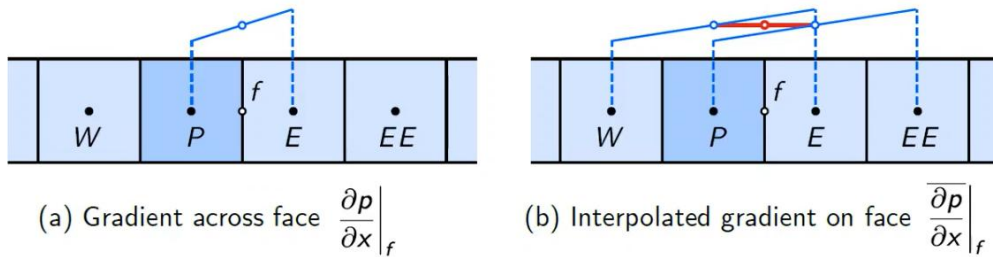
The Implementation

Now all of this is good and well but how do we actually implement it? It is literally like ten lines of code.

Think about everything we have discussed. Where would we put this interpolation? Well we know we need to replace a linear interpolation of the pressure gradient in the pressure correction step, so go to that place in the code. Specifically, we are going to replace the face gradient of the predicted velocity (U_f^*) with Equation (23). Simple as that.

$$U_f^* = [l_x U_p^* + (1 - l_x) U_N^*] + \left[l_x \left(\frac{1}{a_p} \nabla p_p \right) + (1 - l_x) \left(\frac{1}{a_N} \nabla p_N \right) \right] - \left[\frac{1}{a_f} \nabla p_f \right] \quad (23)$$

We already have all of these terms. The l_x comes from geometry. The a_p comes from the discretized momentum equation matrix. The U_p^* and U_N^* are values we have solved for in the previous step for the predicted velocity. The ∇p_f can be found from taking the difference between the cell center pressure values of the cells touching the face as (a) in the diagram below. The only kind of ambiguous part is ∇p_p and ∇p_N . Recall what that entire second bracketed term was. It was the pressure gradient contribution from the linear interpolation that we algebraically removed. That is to say, it is the linearly interpolated pressure gradient so ∇p_p is the pressure gradient determined from its neighboring cells and ∇p_N is the pressure gradient from its respective neighboring cells. The diagram below is shown again for clarity as the second bracketed term is represented by (b).



Long story short, everything can be explicitly evaluated. We have all the information so this is quite simple to implement.

Now what about at the boundaries? Luckily, we don't have to change anything there. This is because the problem came from the fact that the pressure gradient term was linearly interpolated. However, at the boundaries, considering the flux through the boundary face, the values at the face are fixed either through a Dirichlet or Neumann BC. That is to say, we know the value at the cell face so there is no interpolation there so the problem that necessitated Rhie & Chow interpolation is not present. Makes my life easier... This is the end by the way. No conclusion. No summary. Just well wishes.

Notes Thereafter

Upon implementation it has become clear that the Rhie and Chow interpolation is not only needed for the predicted velocity of the pressure correction Poisson equation but EVERYWHERE.

By everywhere I mean that in the discretization of the momentum equation and calculation of the continuity residual, you (or more so I should have) been using the Rhie and Chow interpolation...

Consider this... I can have the same exact solution at the cell centers for my collocated grid but have different values at the faces if I use a linear interpolation versus a Rhie and Chow interpolation.

So, if I was computing the continuity residual ($\nabla \cdot U$) using a linear interpolation and the divergence of the intermediate velocity ($\nabla \cdot U^*$) with a Rhie and Chow interpolation and linear interpolation for the velocity gradient needed in the discretization of the diffusive flux (∇U) this is inconsistent *even if the cell-center values are correct*. For all those terms, the face values are needed so interpolation choice matters and it matters that we are consistent! Or else, you may still get checkerboard oscillations in your solution ... just maybe not as severe. Even though the continuity residual and diffusive flux discretization does not result in checkerboarding (i.e. the odd-even decoupling), you need it for consistency in your calculations. Maybe not *need*, but you *should* have it so velocity being used is consistent across calculations.

This is something that actually got me stuck for a bit but it helped me to realize that Rhie and Chow sounds fancy but it is just a method for interpolation for velocity. EVERYWHERE you interpolate velocity (which I pointed out in the first sentence of the last paragraph), you should use a Rhie and Chow interpolation even if it is not going to cause decoupling of the solution. You need to maintain consistency.